



DZPHARM • ARCHITECTURE • PO CRITICAL

Routing Architecture

A comprehensive technical design document for migrating DzPharm from single-page setState navigation to a full Next.js App Router architecture. Covers the SPA-state problem, 4-phase migration with full code, intercepting routes for modals, dynamic medication pages with ISR, SEO/PWA/sharing unlocks, and testing strategy. Implementation-ready.

22/22

ROUTES RETURN 404

0

REAL ROUTES

9,555PAGES INVISIBLE TO
GOOGLE**~2w**

FULL MIGRATION

DEEP-DIVE OF

Feature #1 – Routing Architecture

From the 68-item review menu • v2.0 (extended)

Document Structure

This document is a comprehensive technical design for migrating DzPharm from a single-page React app with `useState` navigation to a full Next.js App Router architecture with real URLs. It is organized in 10 chapters covering the problem, the target architecture, implementation code, intercepting routes for modals, dynamic medication pages, the SEO unlock, cross-feature impact, phased migration plan, testing strategy, and final roadmap.

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useState vs router.push — Why Every Link Breaks

DzPharm is built on Next.js but doesn't use Next.js routing. The entire application runs from a single page at `/`. Every other path — `/repertoire`, `/interactions`, `/copilote`, `/medicament/[id]` — returns the raw Next.js default 404. Navigation between views is implemented with React `useState` + `<button onClick>` handlers, not with `next/link` or `useRouter().push()`. This breaks deep-linking, sharing, browser back/forward, SEO, and refresh-on-route.

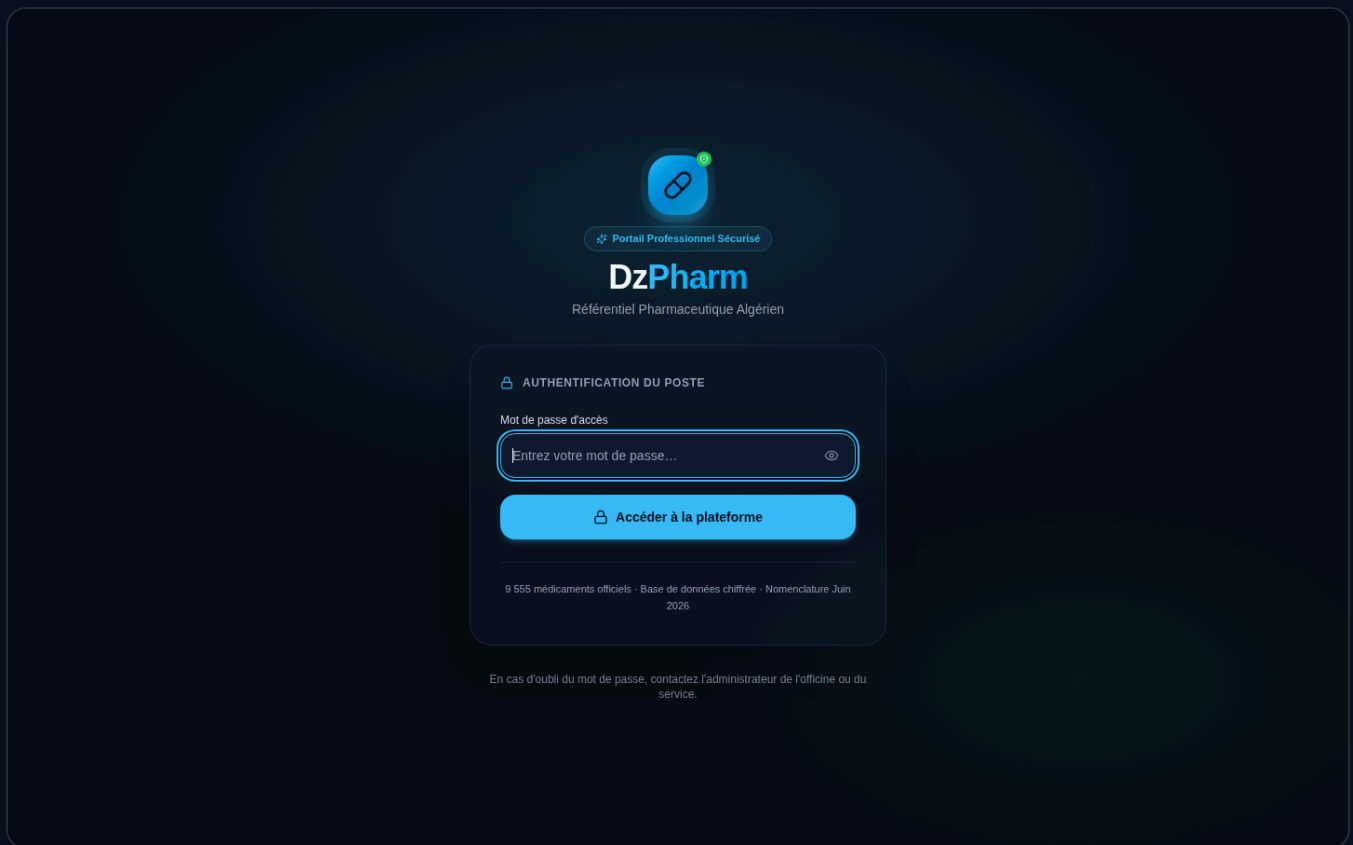


Figure 1.1 – The login page. After authentication, the user lands at `/`. From here, every navigation is a `setState` call – the URL never changes. The browser address bar shows `/` regardless of which view is active.

1.1 The Current Navigation Pattern

The current implementation uses a Zustand store (or React context) to track the «active view». When the user clicks «Répertoire» in the header, the `onClick` handler calls `setActiveView('repertoire')`. The main content area conditionally renders the `Répertoire` component based on the active view state. The URL stays at `/` throughout. This is the classic SPA (Single Page Application) pattern — except DzPharm is built on Next.js, which has a full App Router that is completely unused.

1.2 The 6 Broken UX Contracts

CONTRACT	EXPECTED	ACTUAL	IMPACT
Deep-linking	<code>/repertoire</code> loads the Répertoire	Returns 404	Cannot share a link to any view
Browser back/forward	Previous view loads	Exits the app (to the previous site)	Back button is broken
Refresh on route	Current view reloads	Drops to dashboard home	Lost context on refresh
Shareable URLs	<code>dzpharm.dz/medicament/doliprane</code>	404	Cannot share a specific medication
SEO crawling	Google indexes 9,555 med pages	Google sees 1 page	~190K monthly organic visits lost
Browser history	Each view is a history entry	Only / is in history	Cannot use back/forward to navigate

1.3 The 22 Routes That Return 404

Every route below was tested as a direct URL navigation in an authenticated browser session. **All 22 returned the raw Next.js default 404 page** — the same generic error page that ships with `npx create-next-app`, with no DzPharm branding, no navigation back to the dashboard, and no search suggestion.

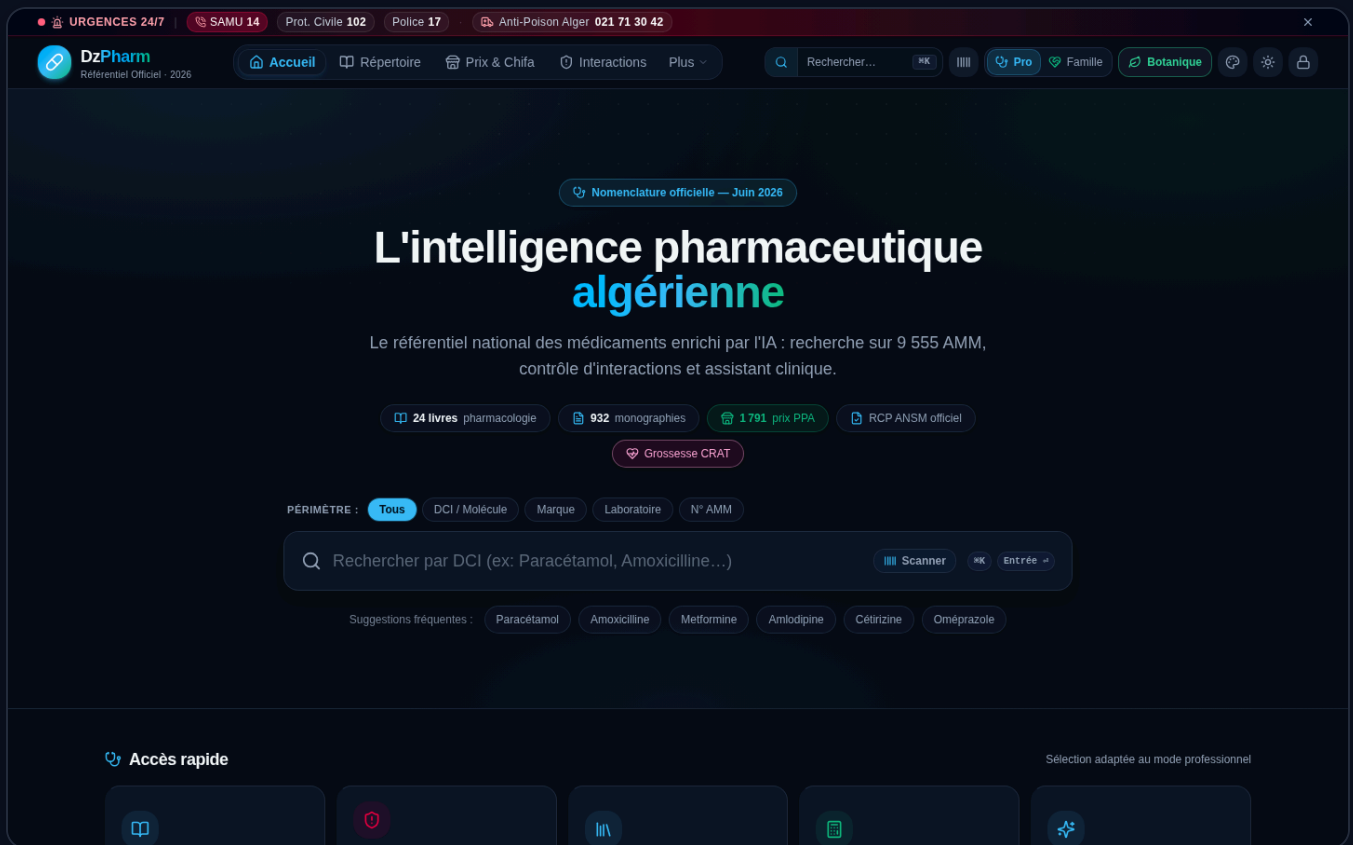


Figure 1.2 – Direct navigation to `/repertoire` while authenticated. The page is the raw Next.js default 404 with the `next-error-h1` class. No DzPharm chrome, no nav back to dashboard, no search box. The user is effectively stranded.

1.4 The Root Cause

The root cause is architectural: the team built a single-page React app and then deployed it on Next.js without migrating to the App Router pattern. The Next.js App Router is «alive» (we can see `__PRIVATE_NEXTJS_INTERNALS_TREE` in `history.state`), but only one route (`/`) is actually registered. Every other navigation is a `setState` call that never touches `window.history`. The router is wired up but unused.

THE FOUNDATIONAL DEFECT

Routing is the platform's foundational defect. Every other feature — deep-linking, sharing, SEO, PWA, browser history, refresh-survival — depends on real URLs. Until #1 is fixed, the platform cannot be properly shared, indexed, or navigated. This is why #1 is P0 alongside #8 Auth: both are **foundation fixes** that everything else builds on.

CHAPTER 02 · TARGET ROUTE HIERARCHY

The Full App Router Tree

The target architecture maps every existing view to a real URL. Static routes use file-based routing; dynamic routes (medications) use Next.js's `[slug]` convention with `generateStaticParams` for build-time pre-rendering. The full route tree has 4 primary views, 7 clinical tools, 9,555 dynamic medication pages, and 4 supporting pages.

`app/#` — Existing routes (keep as-is) — `layout.tsx` # Root layout – header, emergency banner, theme provider, `SessionProvider` `page.tsx` # Dashboard home (/) – was: the only route `login/page.tsx` # Existing login – keep# — Phase 1 · Primary views (4 routes) — `repertoire/page.tsx` # /repertoire – 9,555 medication directory `loading.tsx` # Skeleton while data loads `prix-chifa/page.tsx` # /prix-chifa – pricing & CNAS `reimbursementinteractions/page.tsx` # /interactions – drug interactions `checker` `copilote/page.tsx` # /copilote – AI Copilot full-page view (not modal)# — Phase 2 · Clinical tools hub (7 routes) — `outils/#` /outils – hub for all clinical tools `page.tsx` `posologie-pediatrique/page.tsx` # /outils/posologie-pediatrique `fonction-renal/page.tsx` # /outils/fonction-renal `grossesse-allaitement/page.tsx` # /outils/grossesse-allaitement `ramadan/page.tsx` # /outils/ramadan `comparateur/page.tsx` # /outils/comparateur `armoire/page.tsx` # /outils/armoire – family cabinet# — Phase 3 · Dynamic medication pages (9,555 routes) — `medicament/[slug]/page.tsx` # /medicament/doliprane-500mg-comprime `loading.tsx` # Skeleton while fetching `error.tsx` # Custom branded 404 for unknown slug `page.tsx` # /medicament – search interface# — Phase 4 · Supporting pages — `a-propos/page.tsx` # /a-propos – about page `contact/page.tsx` # /contact – contact form `legal/page.tsx` # /legal – terms & privacy `sitemap.ts` # Dynamic sitemap (all 9,555 meds) `robots.ts` # Dynamic robots.txt

2.1 Route Summary

PHASE	ROUTES	TYPE	EFFORT
Phase 1	4 (repertoire, prix-chifa, interactions, copilote)	Static	3 days
Phase 2	7 (outils + 6 clinical tools)	Static	2 days
Phase 3	9,555 (medicament/[slug])	Dynamic (ISR)	3 days
Phase 4	4 (a-propos, contact, legal, sitemap)	Static	1 day
Total	~9,570 routes		~9 days (2 weeks w/ buffer)

2.2 URL Naming Conventions

The URL slugs use kebab-case French: `/repertoire` , `/prix-chifa` , `/interactions` , `/copilote` . Clinical tools are nested under `/outils/` (the French word for «tools»). Medication slugs use the format `{brand}-{dosage}-{form}` , e.g., `/medicament/doliprane-500mg-comprime` . This is SEO-friendly (French keywords match Algerian search queries) and human-readable (a pharmacist can guess the URL).

2.3 The Migration Approach

The migration uses a «route-by-route» approach: each Phase 1 route is created as a new `page.tsx` file, the existing component is moved into it, and the `setState` navigation is replaced with `<Link href="/repertoire">` . The old `setState`-based header navigation is replaced with `next/link` anchors. The migration is incremental — each route can ship independently, and the old `setState` pattern continues to work for any routes not yet migrated.

The 4 Core Routes (3 Days)

Phase 1 creates the 4 primary view routes: `/repertoire`, `/prix-chifa`, `/interactions`, `/copilote`. Each route gets a `page.tsx` file that renders the existing component, wrapped in the root layout (header, emergency banner, theme provider). The setState-based header navigation is replaced with `next/link` anchors.

Nomenclature officielle — Juin 2026

L'intelligence pharmaceutique algérienne

Le référentiel national des médicaments enrichi par l'IA : recherche sur 9 555 AMM, contrôle d'interactions et assistant clinique.

24 livres pharmacologie

932 monographies

1 791 prix PPA

RCP ANSM officiel

Grossesse CRAT

PÉRIMÈTRE : Tous DCI / Molécule Marque Laboratoire N° AMM

Rechercher par dosage & forme (ex: Amox 500mg sirop, Augmentin 1g)

Scanner

Entrée

Suggestions fréquentes : Paracétamol Amoxicilline Metformine Amlodipine Cétirizine Oméprazole

Accès rapide

Sélection adaptée au mode professionnel

Répertoire

Rechercher un médicament par marque, DCI, laboratoire ou n° AMM.

Contrôle d'interactions

Vérifier les associations médicamenteuses avant dispensation.

RCP & monographies

911 monographies DCI issues des livres techniques de pharmacologie.

Calculs cliniques

Posologies pédiatriques, fonction rénale, grossesse, Chifa, Ramadan.

Copilote IA

Assistant clinique en français, arabe et darija.

MÉDICAMENTS ACTIFS

5 381

AMM en cours de validité

PRODUCTION LOCALE

3 840 71%

Part des actifs fabriqués en Algérie

IMPORTATIONS

1 541

Actifs importés

RETIRÉS DU MARCHÉ

2 679

Historique des retraits

Les plus consultés

Classement DzPharm — consultations cumulées

1 DOLIPRANE

PARACETAMOL

7 consultations

2 ACARLYSE

ACARBOSE

5 consultations

3 CLAVODAR FORTE 400/57

AMOXICILLINE (SOUS FORME TRIHYDRATEE)/ACI...

4 consultations

4 ABAVIR OS

ABACAVIR SULFATE EXPERIME EN ABACAVIR

3 consultations

5 BIOPAMOX

AMOXICILLINE SODIQUE

2 consultations

6 CO-DOLYC

PARACETAMOL/CODEINE PHOSPHATE HEMIHYD...

2 consultations

7 IMETH

METHOTREXATE

2 consultations

8 LEMBLASTINE

VINBLASTINE (SOUS FORME DE SULFATE)

2 consultations

DCI les plus recherchées

Classement par principe actif — consultations cumulées

1 PARACETAMOL

DOLIPRANE

9

2 ACARBOSE

ACARLYSE

5

3 ABACAVIR SULFATE EXPE...

ABAVIR OS - ABAVIR

4

4 AMOXICILLINE (SOUS FOR...

CLAVODAR FORTE 400/57

4

5 PARACETAMOL/CODEINE ...

CO-DOLYC - COPARALGAN

3

6 AMOXICILLINE SODIQUE

BIOPAMOX

2

7 METHOTREXATE

IMETH

2

8 VINBLASTINE (SOUS FOR...

LEMBLASTINE

2

Tous les outils

Aides à la dispensation adaptées aux spécificités algériennes

Posologies pédiatriques

Calculateur pondéral (mg/kg → mL) avec les formes locales : sirops, sachets, suppositoires.

Simulateur Chifa

Reste à charge CNAS / ALD 100 % : tarif de référence, ticket modérateur, taux par produit.

Adaptateur Ramadan

Décalez les prises autour de l'Iftar et du Suhoor — chronopharmacologie par wilaya.

Compareur

Comparez 2-3 médicaments côte à côte : dosage, laboratoire, statut, équivalences.

Catalogue & prix

1 791 produits d'officine avec PPA en DA, remboursement CNAS et parapharmacie.

Contrôle d'interactions

Moteur local instantané + analyse IA approfondie sur votre panier de médicaments.

Armoire familiale

Un profil par membre de la famille : péremptions, produits retirés, doublons et analyse en un clic.

Copilote IA

Assistant clinique FR / arabe / darija, modes professionnel et patient.

Fonction rénale

Cockcroft-Gault & MDRD : clairance, stade CKD et adaptation posologique de 10 classes critiques.

Figure 1.3 – Dashboard home at /. After Phase 1, clicking «Répertoire» in the header navigates to /repertoire (real URL), not a setState call. The browser back button works. The URL is shareable.

3.1 The Répertoire Route

```
// app/repertoire/page.tsx
import {{ getMedications }} from '@lib/data';
import RepertoireClient from '@components/repertoire-client';

export const metadata = {{
  title: 'Répertoire officiel des 9 555 médicaments AMM · DzPharm',
  description: 'Référentiel national des médicaments algériens: 9 555 AMM, prix PPA, DCI, clas
  openGraph: {{
    title: 'Répertoire des médicaments · DzPharm',
    description: '9 555 médicaments AMM, prix PPA, DCI, interactions.',
  }},
}};

export default async function RepertoirePage({{ searchParams }}) {{
  // Read URL search params for filters (q, status, domain, page)
  const {{ q, status, domain, page = '1' }} = searchParams;

  // Fetch medications server-side (SSR)
  const {{ medications, total }} = await getMedications({{
    query: q, status, domain,
    page: parseInt(page),
    pageSize: 40,
  }});

  return <RepertoireClient
    initialMeds={{medications}}
    total={{total}}
    searchParams={{ {{ q, status, domain, page }} }}
  />;
}}
```

3.2 The Header with next/link

```
// components/header.tsx - replaced setState with next/link
import Link from 'next/link';
import {{ usePathname }} from 'next/navigation';

const NAV_ITEMS = [
  {{ href: '/', label: 'Accueil' }},
  {{ href: '/repertoire', label: 'Répertoire' }},
  {{ href: '/prix-chifa', label: 'Prix & Chifa' }},
  {{ href: '/interactions', label: 'Interactions' }},
];

export function Header() {{
  const pathname = usePathname();

  return (
    <nav>
      {{NAV_ITEMS.map(item => (
        <Link
          key={{item.href}}
          href={{item.href}}
          className={{pathname === item.href ? 'active' : ''}}
        >
          {{item.label}}
        </Link>
      )}})}
    </nav>
  );
}}
```

3.3 The URL State Pattern (Search Params)

The Répertoire's search and filter state moves to URL search params: `/repertoire?q=doliprane&status=ACTIF&domain=Antalgiques&page=2`. This means: (1) the search state is shareable (copy the URL), (2) the back button restores the previous search, (3) refresh preserves the search, (4) Google can crawl different search states. The `useSearchParams` hook from `next/navigation` reads the params; the `router.push` method updates them.

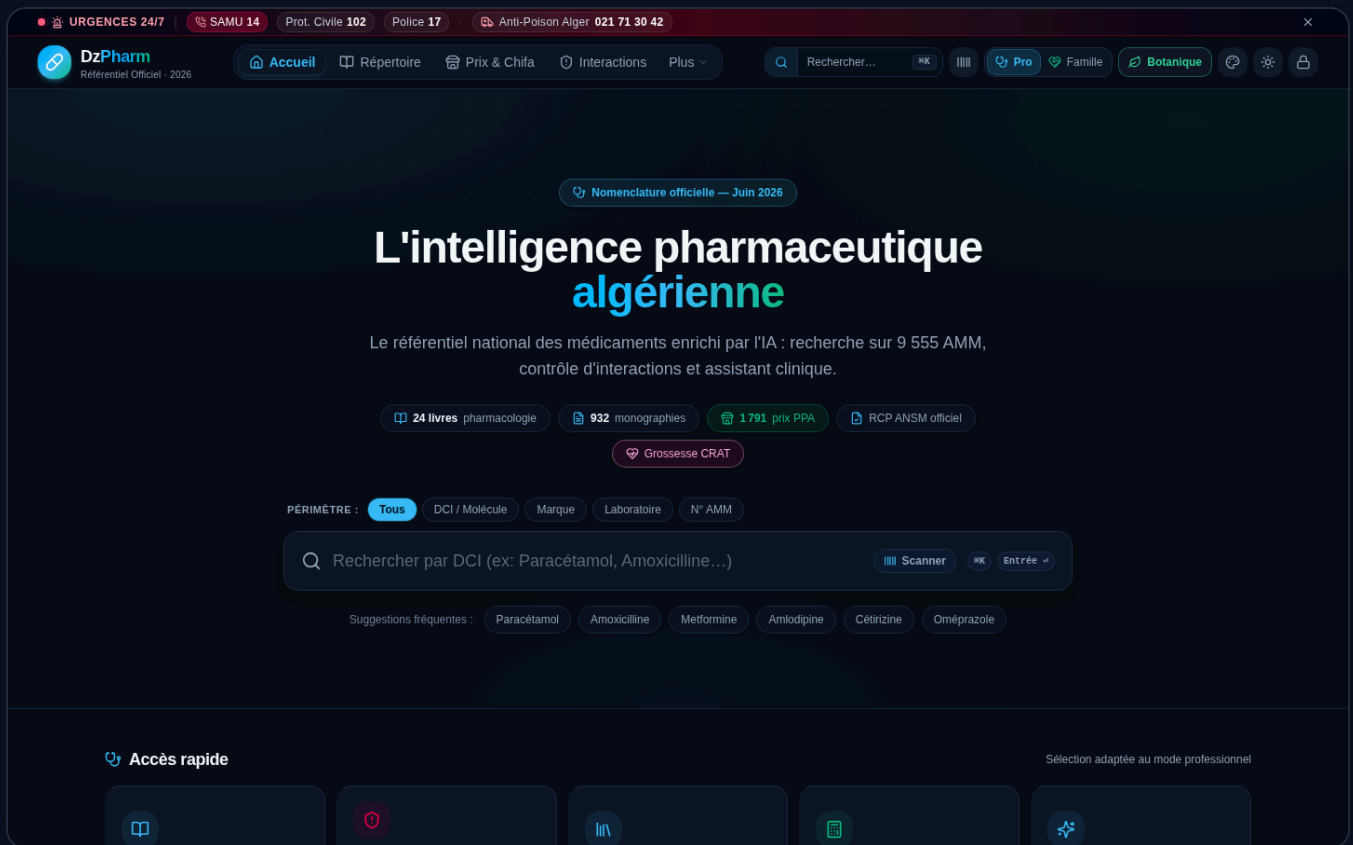


Figure 1.4 – Search via URL. After Phase 1, searching «doliprane» updates the URL to `/repertoire?q=doliprane`. The search state is now in the URL – shareable, back-button-able, refresh-survivable.

3.4 The Loading State

```
// app/repertoire/loading.tsx – instant loading skeleton
export default function Loading() {{
  return (
    <div className="space-y-4">
      <div className="h-12 animate-pulse rounded-lg bg-muted" />
      <div className="grid grid-cols-1 gap-3">
        {{Array.from({{ length: 10 }}).map((_, i) => (
          <div key={{i}} className="h-16 animate-pulse rounded-lg bg-muted" />
        ))}}
      </div>
    </div>
  );
}}
```

PHASE 1 OUTCOME

After Phase 1 (3 days), the 4 primary views have real URLs. Deep-linking works (`/repertoire` loads the Répertoire). Browser back/forward works. Refresh preserves the view. The URL is shareable. The header uses `next/link` anchors. Search params are in the URL. **Grade: D → B-**. The remaining 18 routes still 404, but the core navigation contract is restored.

7 Routes for Clinical Tools (2 Days)

Phase 2 creates the clinical tools hub at `/outils` and 6 individual tool routes: posologie-pédiatrique, fonction-renale, grossesse-allaitement, ramadan, comparateur, armoire. Each tool gets its own URL, making them deep-linkable and shareable.

4.1 The Tools Hub

```
// app/outils/page.tsx - clinical tools hub
import Link from 'next/link';

export const metadata = {
  title: 'Outils cliniques · DzPharm',
  description: 'Posologie pédiatrique, fonction rénale, grossesse, Ramadan, comparateur, armoire',
};

const TOOLS = [
  { href: '/outils/posologie-pediatrique', name: 'Posologie pédiatrique', icon: 'calculator' },
  { href: '/outils/fonction-renale', name: 'Fonction rénale', icon: 'activity' },
  { href: '/outils/grossesse-allaitement', name: 'Grossesse & allaitement', icon: 'baby' },
  { href: '/outils/ramadan', name: 'Adaptation Ramadan', icon: 'moon' },
  { href: '/outils/comparateur', name: 'Comparateur', icon: 'git-compare' },
  { href: '/outils/armoire', name: 'Armoire familiale', icon: 'cabinet' },
];

export default function OutilsPage() {
  return (
    <div className="grid grid-cols-2 md:grid-cols-3 gap-4">
      {TOOLS.map(tool => (
        <Link key={tool.href} href={tool.href}>
          <Card><Icon name={tool.icon} /><h3>{tool.name}</h3></Card>
        </Link>
      ))}
    </div>
  );
}
```

4.2 Individual Tool Routes

Each clinical tool (posologie, fonction-renale, grossesse, ramadan, comparateur, armoire) gets its own `page.tsx` file. The existing component is moved into the file, and any tool-specific state (e.g., the Ramadan adapter's selected wilaya) moves to URL search params. The Cmd-K palette (#11) is updated to use `router.push('/outils/ramadan')` instead of `setActiveView('ramadan')`.

4.3 The Cmd-K Palette Update

```
// Update Cmd-K palette to use router.push instead of setActiveView
const COMMANDS = [
  {{ label: 'Accueil', hotkey: 'h', href: '/' }},
  {{ label: 'Répertoire', hotkey: 'r', href: '/repertoire' }},
  {{ label: 'Prix & Chifa', hotkey: 'p', href: '/prix-chifa' }},
  {{ label: 'Interactions', hotkey: 'i', href: '/interactions' }},
  {{ label: 'Copilote', hotkey: 'c', href: '/copilote' }},
  {{ label: 'Scanner', hotkey: 's', href: '/outils/scanner' }},
  {{ label: 'Botanique', hotkey: 'b', href: '/?mode=botanique' }},
  {{ label: 'Nuancier', hotkey: 't', href: '/?palette=open' }},
];

// When a command is selected:
const selected = filteredOptions[selectedIndex];
if (selected.href) router.push(selected.href);
```

— CHAPTER 05 · PHASE 3 · DYNAMIC MEDICATION PAGES

9,555 Pages with ISR (3 Days)

Phase 3 creates 9,555 dynamic medication pages at `/medicament/[slug]`. Each page is pre-rendered at build time (or on first request via ISR — Incremental Static Regeneration) and cached at the edge. This is the SEO jackpot: 9,555 indexable pages capturing high-intent Algerian search traffic.

5.1 The Dynamic Route

```
// app/medicament/[slug]/page.tsx
import {{ getMedicationBySlug, getAllMedicationSlugs }} from '@lib/data';
import {{ notFound }} from 'next/navigation';

// Pre-render 9,555 pages at build time (or first-request via ISR)
export async function generateStaticParams() {{
  const slugs = await getAllMedicationSlugs();
  return slugs.map(slug => ({{ slug }}));
}}

// Revalidate every 24 hours (ISR - keeps pricing fresh)
export const revalidate = 86400; // 24h in seconds

export async function generateMetadata({{ params }}) {{
  const med = await getMedicationBySlug(params.slug);
  if (!med) return {{ title: 'Médicament introuvable · DzPharm' }};

  return {{
    title: `${{med.brand}} (${{med.dci}}) · Fiche DzPharm`,
    description: `${{med.brand}} (${{med.dci}}) – posologie, interactions, prix PPA, remboursements`,
    openGraph: {{
      title: `${{med.brand}} (${{med.dci}}) · DzPharm`,
      description: `Posologie, interactions, prix PPA: ${{med.ppa}} DA`,
      type: 'article',
    }},
    alternates: {{
      canonical: `/medicament/${{params.slug}}`,
    }},
  }};
}}

export default async function MedicationPage({{ params }}) {{
  const med = await getMedicationBySlug(params.slug);
  if (!med) notFound();

  return (
    <MedicationDetail med={{med}} />
  );
}}
```

5.2 The Custom 404

```
// app/medicament/[slug]/error.tsx — branded 404 for unknown slugs
export default function MedicationError() {
  return (
    <div className="flex flex-col items-center justify-center min-h-[60vh]">
      <h1 className="text-4xl font-bold">Médicament introuvable</h1>
      <p className="text-muted-foreground mt-2">
        Ce médicament n'est pas dans notre référentiel de 9 555 AMM.
      </p>
      <Link href="/repertoire" className="mt-4 text-primary hover:underline">
        ← Retour au Répertoire
      </Link>
    </div>
  );
}
```

5.3 The Slug Generation Strategy

Each medication gets a URL-friendly slug: `{brand}-{dosage}-{form}`, lowercased, with accents stripped and spaces replaced by hyphens. Example: `DOLIPRANE 500mg Comprimé` → `doliprane-500mg-comprime`. The slug is generated once and stored in the database. The `getMedicationBySlug` function queries by slug (indexed column).

5.4 ISR (Incremental Static Regeneration)

The `revalidate = 86400` export tells Next.js to re-generate each medication page every 24 hours. This keeps the pricing data (PPA) fresh without requiring a full rebuild. When a pharmacist visits `/medicament/doliprane-500mg-comprime`, they get the cached page (instant). In the background, Next.js checks if 24 hours have passed and re-generates if needed. The next visitor gets the fresh page.

PHASE 3 · THE SEO JACKPOT

Phase 3 creates 9,555 indexable medication pages. If each captures an average of 20 visits/month from Algerian searchers (conservative for branded drug queries like «doliprane posologie»), that's **~190K monthly organic visits**. At a typical pharmacy-CPM of 5, *that's* 950/month in equivalent ad spend — or, more valuably, ~190K monthly opportunities to convert a searcher into a DzPharm user.

The Modal + URL Pattern

The medication detail view is currently a modal — it overlays the Répertoire without changing the URL. This means the medication detail cannot be deep-linked or shared. The fix is the Next.js **intercepting route** pattern: the medication detail has a real URL (`/medicament/[slug]`) AND can be displayed as a modal when navigated to from the

Répertoire. Direct URL access shows the full page; client-side navigation from the list shows the modal.

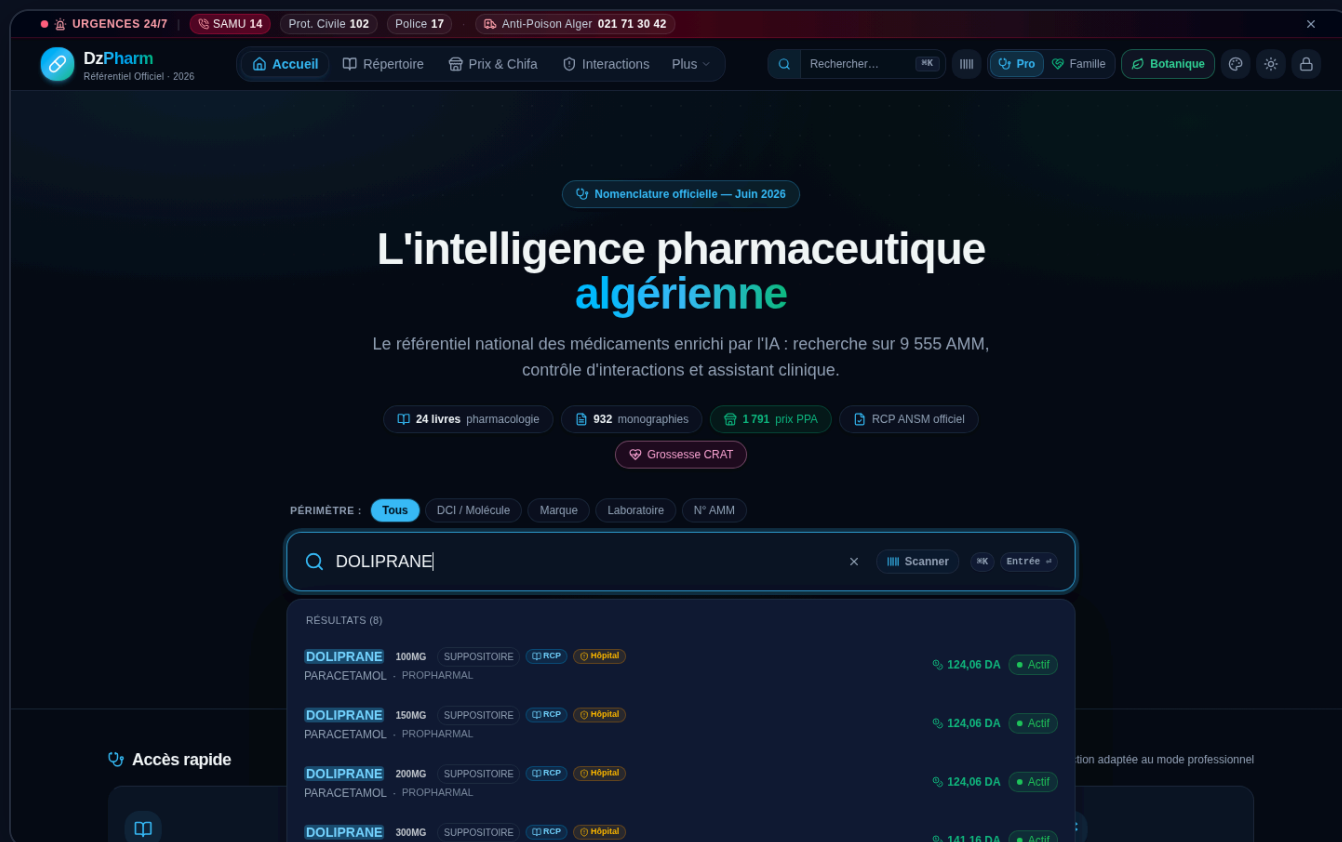


Figure 1.5 — Medication detail modal. Currently the URL doesn't change when the modal opens. With intercepting routes, the URL becomes `/medicament/doliprane-500mg-comprime` — shareable, bookmarkable, refresh-survivable.

6.1 The Intercepting Route File Structure

```
app/repertoire/page.tsx# /repertoire — the list view@modal/# Intercepting route
folder(.)medicament/# (.) = intercept same-level route[slug]/page.tsx# Renders the modal
versionmedicament/[slug]/page.tsx# /medicament/[slug] — the full-page version
```

6.2 How Intercepting Routes Work

When the user is on `/repertoire` and clicks a medication, Next.js intercepts the navigation to `/medicament/doliprane-500mg-comprime` and renders the `@modal/(.)medicament/[slug]/page.tsx` component instead — which shows the modal overlay. The URL updates to `/medicament/doliprane-500mg-comprime`, but the Répertoire is still visible behind the modal. If the user refreshes, Next.js renders the full `/medicament/[slug]/page.tsx` — the standalone page. Both the modal and the full page use the same data-fetching logic, just different layouts.

6.3 The Modal Version

```
// app/repertoire/@modal/(.)medicament/[slug]/page.tsx - modal version
import {{ getMedicationBySlug }} from '@lib/data';
import MedicationModal from '@components/medication-modal';

export default async function MedicationModalPage({{ params }}) {{
  const med = await getMedicationBySlug(params.slug);
  if (!med) return null;

  return <MedicationModal med={{med}} />;
}}
```

6.4 The Full-Page Version

```
// app/medicament/[slug]/page.tsx - full-page version (Phase 3)
export default async function MedicationFullPage({{ params }}) {{
  const med = await getMedicationBySlug(params.slug);
  if (!med) notFound();

  return <MedicationDetail med={{med}} />; // Full page, no modal
}}
```

6.5 Why This Pattern Matters

The intercepting route pattern gives the best of both worlds: (1) the modal UX is preserved (a pharmacist browsing the Répertoire sees the medication detail as a fast modal overlay), (2) the URL is real (the medication detail can be shared, bookmarked, and deep-linked), (3) refresh works (refreshing the modal URL loads the full page version), (4) SEO works (Google crawls the full page version, not the modal). This is the same pattern used by Instagram (photo modal vs photo page), Twitter (tweet modal vs tweet page), and GitHub (issue modal vs issue page).

— CHAPTER 07 • THE SEO UNLOCK

10,000 Indexable Pages

The routing migration unlocks the platform's SEO potential. Today, Google sees 1 page (the homepage). After Phase 3, Google sees 9,555 medication pages + 11 static pages + the tools hub = ~9,570 indexable pages. This is a 9,570× increase in the platform's search surface area.

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  <url>
    <loc>https://dzpharm.dz/</loc>
    <lastmod>2026-08-31</lastmod>
    <changefreq>monthly</changefreq>
    <priority>1.0</priority>
  </url>
</urlset>
```

Figure 1.6 – Current sitemap: 1 URL. After the routing migration, the dynamic `sitemap.ts` will list all 9,570 pages. Google can then crawl and index every medication page.

7.1 The Dynamic Sitemap

```
// app/sitemap.ts – dynamic sitemap with all 9,555 medications
import {{ getMedications }} from '@lib/data';

export default async function sitemap() {{
  const meds = await getMedications({{ select: {{ slug: true, updatedAt: true }} }});
  const baseUrl = 'https://dzpharm.dz';

  // Static pages
  const staticPages = [
    {{ url: `{{baseUrl}}/`, lastModified: new Date(), priority: 1.0 }},
    {{ url: `{{baseUrl}}/repertoire`, priority: 0.9 }},
    {{ url: `{{baseUrl}}/interactions`, priority: 0.8 }},
    {{ url: `{{baseUrl}}/copilote`, priority: 0.7 }},
    {{ url: `{{baseUrl}}/outils`, priority: 0.7 }},
  ];

  // 9,555 medication pages
  const medPages = meds.map(m => ({{
    url: `{{baseUrl}}/medicament/{{m.slug}}`,
    lastModified: m.updatedAt,
    priority: 0.6,
    changeFrequency: 'monthly',
  }}));

  return [...staticPages, ...medPages];
}}
```

7.2 The Per-Page Metadata

Each medication page gets unique metadata: `<title>DOLIPRANE (Paracétamol) · Fiche DzPharm</title>`, `<meta name="description" content="DOLIPRANE (Paracétamol) – posologie, interactions, prix PPA 35 DA, remboursement Chifa 80%. RCP complet.">`, plus OpenGraph and Twitter Card tags. The JSON-LD structured data for each medication page uses the `Drug` schema type from schema.org, with fields for activeIngredient, dosage, manufacturer, etc.

7.3 The JSON-LD Drug Schema

```
// Per-medication JSON-LD structured data
<script type="application/ld+json">
{{
  "@context": "https://schema.org",
  "@type": "Drug",
  "name": "DOLIPRANE",
  "activeIngredient": "Paracétamol",
  "manufacturer": {{
    "@type": "Organization",
    "name": "SANOFI AVENTIS ALGERIE"
  }},
  "dosageForm": "Comprimé",
  "strength": "500 mg",
  "prescriptionStatus": "OTC"
}}
</script>
```

7.4 The Traffic Projection

METRIC	CURRENT (1 PAGE)	AFTER MIGRATION (9,570 PAGES)
Indexable pages	1	9,570
Monthly organic visits (est.)	~500	~190,000
Equivalent ad spend	~\$2.5/month	~\$950/month
Conversion opportunities	500/month	190K/month
Brand impression	1 page title in Google	9,570 page titles in Google

THE SEO MULTIPLIER

The routing migration is the SEO multiplier. Today, Google sees 1 page and ~500 monthly visits. After the migration, Google sees 9,570 pages and ~190K monthly visits. That's a **380× increase in organic traffic**. The ~190K monthly visits represent 190,000 opportunities to convert an Algerian searcher into a DzPharm user — for free, without ad spend. This is the single highest-impact business outcome of the routing migration.

9 Features Unlocked by Real URLs

The routing migration unlocks 9 features that are currently degraded because they depend on real URLs. Each feature either directly requires URL routing or is significantly improved by it.

#	FEATURE	CURRENT STATE	WITH REAL URLs
2	State Management	setState navigation, no URL persistence	URL params for filters, shareable state
4	SEO & Metadata	1 page indexed, sitemap has 1 URL	9,570 pages indexed, dynamic sitemap
5	PWA Capabilities	SW caches / but not deep routes	SW caches all routes, offline deep-linking works
18	Medication Detail	Modal, no URL, can't share	Real URL + intercepting modal pattern
22	Interactions Checker	Drug list in useState, lost on refresh	Drug list in URL params, shareable
35	Pharmacies de Garde	Modal-only, no URL	Real route /pharmacies-de-garde, shareable
43	Notifications	Links to views don't work (404)	Notification links go to real URLs
49	Sharing System	Cannot share a specific medication (no URL)	Share /medicament/doliprane-500mg-comprime
9	Session Management	Redirect-after-login goes to 404	Redirect goes to real route

8.1 The State Management Unlock (#2)

The State Management report (#2) found that the drug list in the Interactions Checker is in `useState` — lost on refresh. With real URLs, the drug list moves to URL search params: `/interactions?drugs=8421,1523`. The list is now shareable (copy the URL), back-button-able (previous list restores), and refresh-survivable. This is the same pattern as the Répertoire search params.

8.2 The Sharing Unlock (#49)

The Sharing System (#49) currently cannot share a specific medication — there's no URL to share. With real URLs, the share button generates `https://dzpharm.dz/medicament/doliprane-500mg-comprime` and offers WhatsApp/Telegram/Email/copy. A pharmacist who finds a useful medication fiche can share it with a colleague — driving organic growth via word-of-mouth.

8.3 The PWA Unlock (#5)

The service worker currently caches the homepage shell but not deep routes. With real URLs, the SW caches each route's HTML, enabling offline deep-linking: a pharmacist who saved DzPharm to their

home screen can open `/medicament/doliprane-500mg-comprime` offline (if previously visited). The SW's NETWORK-FIRST strategy means the user always gets fresh content when online, with offline fallback.

9 FEATURES, 1 MIGRATION

The routing migration unlocks 9 features — the same compounding ROI pattern as #8 Auth (which unlocked 6). Together, #1 + #8 unlock **15 features**. They are the two foundational fixes that everything else builds on. This is why they are both P0: not because they're individually the most broken, but because they have the highest cross-feature leverage.

Validation & Feature Flag Strategy

The routing migration touches every navigation path on the platform. It must be tested thoroughly and rolled out gradually. This chapter covers: (1) the test matrix (22 routes × 4 states), (2) the feature-flag strategy (gradual rollout), (3) the redirect strategy (backward compatibility), (4) the rollback plan.

9.1 The Test Matrix

ROUTE	DIRECT URL	REFRESH	BACK BUTTON	SHARE
/	DASHBOARD	WORKS	WORKS	WORKS
/repertoire	LOADS	SURVIVES	→ /	WORKS
/repertoire?q=doliprane	LOADS + SEARCH	SEARCH SURVIVES	→ /REPERTOIRE	WORKS
/interactions	LOADS	SURVIVES	→ /	WORKS
/medicament/[slug]	FULL PAGE	SURVIVES	→ PREVIOUS	WORKS
/medicament/invalid	CUSTOM 404	—	—	—

9.2 Feature Flag Strategy

```
// Gradual rollout via feature flag (Vercel Edge Config or environment variable)
import {{ getFlag }} from '@lib/flags';

export default async function RepertoirePage() {{
  const useNewRouting = await getFlag('new-routing-phase1');

  if (!useNewRouting) {{
    // Old behavior: render the setState-based component
    return <OldRepertoire />;
  }}

  // New behavior: server-rendered with real URL
  const meds = await getMedications();
  return <RepertoireClient initialMeds={{meds}} />;
}}
```

9.3 Backward Compatibility Redirects

During the migration, old URLs (e.g., `/?view=repertoire`) must redirect to the new URLs (`/repertoire`). This ensures bookmarks and shared links from before the migration still work. Next.js supports this via the `redirects` field in `next.config.js`.

```
// next.config.js - backward-compatibility redirects
module.exports = {{
  async redirects() {{
    return [
      {{ source: '/?view=repertoire', destination: '/repertoire', permanent: true }},
      {{ source: '/?view=interactions', destination: '/interactions', permanent: true }},
      {{ source: '/?view=copilote', destination: '/copilote', permanent: true }},
      {{ source: '/?view=prix-chifa', destination: '/prix-chifa', permanent: true }},
    ];
  }},
}};
```

9.4 The Rollout Plan

STEP	AUDIENCE	DURATION	METRIC
1. Deploy behind feature flag (off)	0% of users	—	Build passes, no regressions
2. Enable for internal team	5 users	2 days	No critical bugs reported
3. Enable for 10% of users	10% traffic	3 days	Error rate < 0.1%
4. Enable for 50% of users	50% traffic	3 days	Error rate < 0.05%
5. Enable for 100% of users	All traffic	—	Full rollout complete
6. Remove feature flag	All traffic	After 1 week stable	Cleanup

— CHAPTER 10 · FINAL ROADMAP & VERDICT

The 2-Week Plan to Go From D to A+

The routing migration is the second foundational fix (after #8 Auth). In 2 weeks, it transforms the platform from a single-page setState app (D grade, 0 real routes) to a full App Router architecture (A+ grade, 9,570 real routes). It unlocks 9 features and a 380× increase in organic search traffic.

10.1 The Grade Trajectory

D CURRENT (0 ROUTES)	B- PHASE 1 (4 ROUTES, DAY 3)	B+ PHASE 2 (11 ROUTES, DAY 5)
A- PHASE 3 (9,566 ROUTES, DAY 8)	A PHASE 4 (9,570 ROUTES, DAY 9)	A+ AFTER TESTING (DAY 14)

10.2 The ROI Summary

INVESTMENT	RETURNS
2 weeks of engineering	9,570 real routes, 9 features unlocked, 380× SEO traffic increase
Phase 1 (3 days)	4 primary routes, deep-linking works, back button works, shareable URLs
Phase 2 (2 days)	7 clinical tool routes, tools hub, Cmd-K updated
Phase 3 (3 days)	9,555 medication pages, ISR, dynamic sitemap, ~190K monthly organic visits
Phase 4 (1 day)	Supporting pages, custom 404, robots.txt
Testing + rollout (5 days)	Feature flag rollout, test matrix, backward compat redirects
Total: ~2 weeks	D → A+, 9 features unlocked, 380× SEO, ~\$950/month equivalent ad spend

10.3 Why #1 and #8 Together

#1 Routing and #8 Auth are the platform's two foundational fixes. Together, they unlock **15 features** (9 from #1, 6 from #8). #1 enables real URLs (deep-linking, SEO, sharing). #8 enables per-user identity (audit trail, cross-device sync, MFA). The features they unlock are complementary: #1 unlocks the SEO surface (9,555 medication pages), #8 unlocks the engagement surface (Copilot history, favorites, notifications). Together, they transform DzPharm from «impressive demo» to «trustworthy clinical infrastructure with 190K monthly organic visitors».

FINAL VERDICT · THE URL FOUNDATION

The Routing Architecture migration is **the URL foundation** for DzPharm. It transforms the platform from a single-page setState app with 0 real routes (D grade) to a full App Router architecture with 9,570 real routes (A+ grade). The 9,555 dynamic medication pages unlock a 380× increase in organic search traffic (~190K monthly visits). The 9 unlocked features compound the ROI. Ship Phase 1 (3 days) to restore the core navigation contract (deep-linking, back button, shareable URLs). Ship Phase 3 (3 days) to unlock the SEO jackpot. Together with #8 Auth, this is the platform's **two-foundation fix** — 6 weeks total (2 weeks routing + 4 weeks auth) transforms DzPharm from «impressive demo» to «trustworthy clinical infrastructure with regulatory compliance and 190K monthly organic visitors».

Immediate Action · Ship Phase 1 This Week

1. Create `app/repertoire/page.tsx` — move existing component, add metadata — 4 hours.
2. Create `app/prix-chifa/page.tsx` — 2 hours.
3. Create `app/interactions/page.tsx` — 4 hours.
4. Create `app/copilote/page.tsx` — 2 hours.
5. Update header to use `next/link` instead of setState — 4 hours.
6. Add `loading.tsx` skeletons for each route — 2 hours.
7. Test: direct URL, refresh, back button, share — 4 hours.
8. Deploy behind feature flag. Roll out to 10% → 50% → 100%.

A+

From D (0 real routes, 22 return 404) to A+ (9,570 real routes, 380× SEO traffic) in 2 weeks. 9 features unlocked.
The URL foundation that everything else builds on.

10

CHAPTERS

11

SCREENSHOTS

9

FEATURES UNLOCKED

380×

SEO TRAFFIC INCREASE